Spivak Chapter 1 Problems

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1 Problem 1

Prove the following:

1. If $ax = a$ for some number $a \neq 0$, then $x = 1$.
   $ax = a$,
   $a^{-1} \cdot ax = a^{-1} \cdot a$,
   $(a^{-1} \cdot a) \cdot x = a^{-1} \cdot a$,
   $1 \cdot x = 1$, therefore,
   $x = 1$.
2. $x^2 - y^2 = (x - y)(x + y)$.
   $x^2 - y^2 = (x - y)(x + y)$,
   $x^2 - y^2 = x(x + y) - y(x + y)$,
   $x^2 - y^2 = x^2 + xy - xy - y^2$,
   $x^2 - y^2 = x^2 - y^2$.
3. If $x^2 = y^2$, then $x = y$ or $x = -y$.
   $x^2 = y^2$,
   $x \cdot y^{-1} = y \cdot x^{-1}$, thus,
   $x = \pm y$.

2 Problem 5

Prove the following:

1. If $a < b$ and $c < d$, then $a + c < b + d$.
   $a < b$, $c < d$,
   $a + c < b + c$, $b + c < b + d$, therefore,
   $a + c < b + d$.

2. If $a < b$, then $-b < -a$.

   $a - b < 0$,

   $-(a - b) > 0$, thus,

   $-b < -a$.

3. If $a < b$ and $c > d$, then $a - c < b - d$.

   $a < b$, $c > d$,

   $a - c < b - c$, $b - c < b - d$, therefore,

   $a - c < b - d$.

4. If $a < b$ and $c > 0$, then $ac < bc$.

   $b - a \in P$,

   $c \cdot (b - a) \in P$,

   $bc - ac \in P$, i.e.,

   $ac < bc$.

5. If $a < b$ and $c < 0$, then $ac > bc$.

   $b - a \in P$,

   $-1 \cdot c \cdot (b - a) \in P$,

   $c \cdot (a - b) \in P$,

   $ac - bc \in P$,

   $ac > bc$.

6. If $0 < a < 1$, then $a^2 < a$.

   Let $c = a$. Then, by (iv), $a^2 < ba$. For $b = 1$, $a^2 < a$.

7. If $0 \leq a < b$ and $0 \leq c < d$, then $ac < bd$.

   If $0 \leq a < b$, then $0 \leq ac < bc$. In addition, if $0 \leq c < d$, then $0 \leq bc < bd$. Thus, $ac < bd$, assuming $b > 0, d > 0$.

3 Problem 6

1. Prove that if $0 \leq x < y$, then $x^n < y^n$.

   Assume $x^k < y^k$. As given, for $k = 1$ we have $0 \leq x < y$. For the case of $k + 1$, we have $0 \leq x^{k+1} < x \cdot y^k$ and, $0 \leq x \cdot y^k < y^{k+1}$. Thus, $0 \leq x^{k+1} < y^{k+1}$ for all n.

2. Prove that if $x < y$ and n is odd, then $x^n < y^n$.

   For $k = 0$, $x^1 < y^1$, as given. For $x, y \geq 0$ n, as per problem 6(a), $x^n < y^n$. For negative x, y, we have $0 \geq y^n > x^n$. For negative x and positive y, clearly $x < y$, and, since n is odd, $x^n < y^n$.

4 Problem 10

Express each of the following without absolute value signs, treating various cases separately when necessary.

1. $|a+b|-|b|$

$$|a+b|-|b| = \begin{cases} a+b-b, & a+b>0, b>0 \\ a+b+b, & a+b>0, b<0 \\ -(a+b)-b, & a+b<0, b>0 \\ -(a+b)+b, & a+b<0, b<0 \end{cases}$$

2. $|(|x|-1)|$

$$|(|x|-1)| = \begin{cases} x<-1, & -x-1 \\ -1 \leq x < 0, & 1+x \\ 0 < x \leq 1, & 1-x \\ x \geq 1, & x-1 \end{cases}$$

5 Problem 11

Find all numbers x for which

1. $|x-3|=8$.
   $x = 11, -5$.
2. $|x-3|<8$.
   $-5 < x < 11$.
3. $|x+4|<2$.
   $-6 < x < -2$.
4. $|x-1|+|x-2|>1$.
   $x \in (-\infty, 1) \cup (2, \infty)$.

6 Problem 13

The maximum of two numbers x and y is denoted by $max(x,y)$. Thus $max(-1,3) = max(3,3) = 3$ and $max(-1,-4) = max(-4,-1) = -1$. The minimum of x and y is denoted by $min(x,y)$. Prove that

$$max(x,y) = \frac{x+y+|y-x|}{2},$$

$$min(x,y) = \frac{x+y-|y-x|}{2}.$$

Let $max(x,y) = x$. Then

$$x+y+|y-x| = 2x,$$
$$max(x,y) = x = \frac{x+y+|y-x|}{2}.$$

Now let $min(x,y) = x$. Then

$$x+y-|y-x| = 2x,$$
$$min(x,y) = x = \frac{x+y-|y-x|}{2}.$$

7 Problem 20

Prove that if
$$|x-x_0| < \frac{\varepsilon}{2}$$
and
$$|y-y_0| < \frac{\varepsilon}{2},$$
then
$$|(x+y)-(x_0+y_0)| < \varepsilon,$$
$$|(x-y)-(x_0-y_0)| < \varepsilon.$$

We have

$$|(x-x_0)+(y-y_0)| \leq |x-x_0|+|y-y_0| < \frac{\varepsilon}{2}+\frac{\varepsilon}{2}.$$

The desired results follow from application of the triangle inequality.